Fiscal Year:	FY 2012	**Task Last Updated:**	FY 10/20/2011
PI Name:	Catauro, Patricia M.S.		
Project Title:	Comparative Packaging Study		
Division Name:	Human Research		
Program/Discipline:	HUMAN RESEARCH		
Program/Discipline--Element/Subdiscipline:	HUMAN RESEARCH--Space Human Factors Engineering		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	(1) **SHFH**:Space Human Factors & Habitability (archival in 2017)		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
PI Email:	patricia.catauro@nasa.gov, erin.s.connell@nasa.gov	**Fax:**	FY
PI Organization Type:	NASA CENTER	**Phone:**	281.483.3632
Organization Name:	NASA Johnson Space Center/Lockheed Martin		
PI Address 1:	1300 Hercules MC:C09		
PI Address 2:			
PI Web Page:			
City:	Houston	**State:**	TX
Zip Code:	77058	**Congressional District:**	22
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	Directed Research
Start Date:	10/01/2008	**End Date:**	10/01/2011
No. of Post Docs:		**No. of PhD Degrees:**	0
No. of PhD Candidates:		**No. of Master' Degrees:**	0
No. of Master's Candidates:		**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:		**Monitoring Center:**	NASA JSC
Contact Monitor:	Perchonok, Michele	**Contact Phone:**	(281) 483-7632
Contact Email:	michele.perchonok22@nasa.gov		
Flight Program:			
Flight Assignment:	NOTE: End date changed to 10/1/2011 for reporting purposes (Ed., 9/26/2011)		
Key Personnel Changes/Previous PI:	NOTE: PI changed from Michele Perchonok to Patricia Catauro effective FY2009 (10/01/2008); was briefly Thomas Oziomek but changed to Catauro during the same fiscal year, i.e., FY2009 [information per M. Perchonok/JSC]. See Perchonok for FY2008 reporting.		
COI Name (Institution):			
Grant/Contract No.:	Directed Research		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	Advanced Food Technology (AFT) is currently working to design a stable, palatable, and nutritious food supply, to support long-duration missions beyond Low-Earth Orbit (LEO). Identifying high-barrier food packaging is integral to this work, as food quality is greatly reduced upon interaction with environmental moisture and oxygen. This research project evaluated the long-term performance of three packaging materials: a low barrier material, a clear high barrier material, and an opaque high barrier material. One ingredient and two food products, all extremely vulnerable to oxygen- and moisture-induced spoilage, were packaged in the three materials. The quality of these food items with long-term storage at relevant temperature and humidity parameters was examined regularly by instrumental analysis and sensory evaluation. Quality of these food items over time has indicated the relative longevity of each packaging material. Data obtained have suggested the continued need of a metalized overwrap material to ensure the quality and shelf life of foods packaged in clear materials for ISS and long duration spaceflight applications. Critical limits of quality for evaluated food products and recommendations for packaging of NASA food items in long duration missions were identified based on the findings of this study.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	The study compares three commercially available packaging materials. The results may be helpful to food industry. No innovative technologies are being developed.
Task Progress:	Advanced Food Technology (AFT) is currently working to design a stable, palatable, and nutritious food supply to support long-duration missions beyond Low-Earth Orbit (LEO). Identifying high barrier food packaging is integral to this work, because food quality is greatly reduced upon exposure to environmental moisture and oxygen. The Comparative Packaging Study research project evaluated the long-term performance of three packaging materials: a low barrier transparent material, a high-barrier transparent material, and a high barrier opaque material. One ingredient and two food products, all extremely vulnerable to oxygen and moisture induced spoilage, were packaged in the three test materials. The quality of these food items with long term storage at relevant temperature (21-23oC) and humidity (25%RH – 75%RH) parameters was examined regularly for 36 months. During this time, the foods were evaluated for any increases in moisture content, increases in staling aroma compounds, and changes in other food quality indicators. Sensory testing (or taste testing) was also used to assess the quality of the foods over storage time, by having volunteers compare the aging items to control samples. The assessment of the packaged food items over time has indicated the relative longevity of each of the three packaging materials. The opaque material preserved the food quality the longest of the three materials (>36 months), and its success appeared to be unaffected by the different storage conditions studied. After considering all results, AFT researchers were able to establish minimum limits for food quality indicators, which will likely simplify food storage studies for NASA missions in the future. AFT researchers also used the results of this study to make recommendations on the use of the three materials to support current and future NASA missions. Among these recommendations, AFT researchers encouraged the use of the high barrier opaque material to ensure the quality of food for storage on ISS.
Bibliography Type:	Description: (Last Updated: 03/22/2018)